

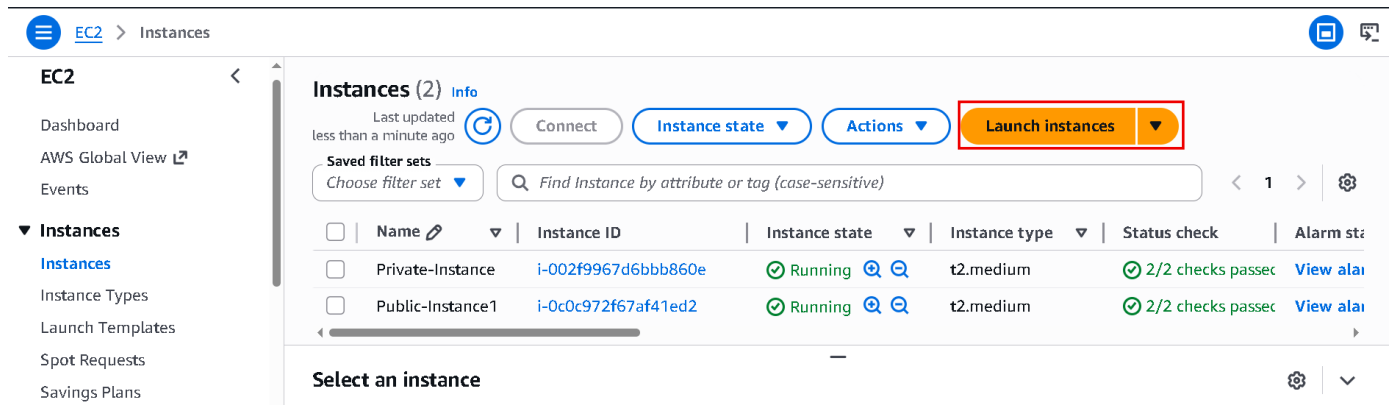
EBS Volume: -

Summary: -

In this lab, I created a Windows EC2 instance and connected it using Remote Desktop. Then I created and attached a new EBS volume for extra storage. After that, I initialized the disk, created a new drive, and increased the storage size when more space was needed

Let's Create: -

- Search EC2 in Aws Console and Open it.
- Now Let's create windows EC2 instance.



The screenshot shows the AWS Management Console for the EC2 service. The left sidebar contains navigation links for EC2, Dashboard, AWS Global View, Events, and a list of instance types and templates. The main content area displays a list of instances. At the top, there are buttons for 'Connect', 'Instance state', 'Actions', and 'Launch instances' (highlighted with a red box). Below these buttons is a search bar and a table of instances. The table has columns for Name, Instance ID, Instance state, Instance type, Status check, and Alarm state. Two instances are listed: 'Private-Instance' and 'Public-Instance1', both in a 'Running' state.

Name	Instance ID	Instance state	Instance type	Status check	Alarm state
Private-Instance	i-002f9967d6bbb860e	Running	t2.medium	2/2 checks passed	View alarm
Public-Instance1	i-0c0c972f67af41ed2	Running	t2.medium	2/2 checks passed	View alarm

- Now opens EC2 in new tab.
- And click Launch instance.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

WindowsInstance

[Add additional tags](#)

- Enter instance name.
- Scroll down.

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

 Search our full catalog including 1000s of application and OS images

Quick Start



Amazon Linux



macOS



Ubuntu



Windows



Red Hat



SUSE



[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

- Choose OS click Browse more AMIs.

≡ [EC2](#) > [Instances](#) > [Launch an instance](#) > AMIs



 Search for an AMI by entering a search term e.g. "Windows"



Quick Start AMIs (45)

Commonly used AMIs

My AMIs (13)

Created by me

AWS Marketplace AMIs (6091)

AWS & trusted third-party AMIs

Community AMIs (500)

Published by anyone



Windows

Free tier eligible

Verified provider

Microsoft Windows Server 2019 Base

ami-08e30c01541e6a4f3 (64-bit (x86))

Microsoft Windows 2019 Datacenter edition.
[English]

Platform: windows
Virtualization: hvm

Root device type: ebs
ENA enabled: Yes

Select

64-bit (x86)

- I am going to use windows 2019 Base.
- Click on select and check first they are Free tier eligible or not.

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

 Search our full catalog including 1000s of application and OS images

[AMI from catalog](#)

[Recents](#)

[My AMIs](#)

[Quick Start](#)

Name

Microsoft Windows Server 2019 Base

Verified provider

Free tier eligible

Description

Microsoft Windows 2019 Datacenter edition. [English]

Microsoft Windows Server 2019 with Desktop Experience Locale English AMI provided by Amazon

Image ID

ami-08e30c01541e6a4f3



[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

- Now scroll down.

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Linux base pricing: 0.0104 USD per Hour

☒ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

- now we are just going to select this t3.micro
- In this we are going to get two virtual CPU and one GB Ram.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Select

 [Create new key pair](#)

- Click Create new key pair.

Create key pair

Key pair name

Key pairs allow you to connect to your instance securely.

demo

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

Cancel

Create key pair

- Enter key name.
- Key pair type as RSA
- And .pem file
- And last click Create key pair
- Now your keypair is downloaded and save it.
- Now scroll down.

▼ Network settings Info

Edit

Network Info

vpc-072c4acd761c3b942

Subnet Info

No preference (Default subnet in any availability zone)

Auto-assign public IP Info

Enable

Additional charges apply when outside of free tier allowance

- Now click Edit.

▼ Network settings [Info](#)

VPC - *required* | [Info](#)

vpc-072c4acd761c3b942
172.31.0.0/16

(default) ▼



Subnet | [Info](#)

subnet-04ffa8e5e9c79ad3d

VPC: vpc-072c4acd761c3b942 Owner: 463646775279

Availability Zone: ap-south-1a (aps1-az1) Zone type: Availability Zone

IP addresses available: 4089 CIDR: 172.31.32.0/20



[Create new subnet ↗](#)

Auto-assign public IP | [Info](#)

Enable ▼

Additional charges [apply](#) when outside of [free tier allowance](#)

- Now Default VPC not change it.
- Select Subnet ab-south-1a because in Mumbai region three availability zones 1a, 1b, 1c if we not choose then aws decide in which availability zones they want to create our EC2 instance.

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - *required*

launch-wizard-80

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and . _ - / 0 # , @ [] + = & ; ! \$ *

Description - *required* | [Info](#)

launch-wizard-80 created 2026-05-19T08:26:27.229Z

- Now here create a security group or you select your created security group also.
- Aws always create a custom security group if you want to change name of security group you can change it.

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 3389, 0.0.0.0/0)

[Remove](#)

Type | [Info](#)

rdp ▼

Protocol | [Info](#)

TCP

Port range | [Info](#)

3389

Source type | [Info](#)

Anywhere ▼

Source | [Info](#)

Add CIDR, prefix list or securi

0.0.0.0/0 ✕

Description - *optional* | [Info](#)

e.g. SSH for admin desktop

- You can add extra permissions in your security group.
- By default for windows rdb permissions is added.
- Now scroll down.

▼ Configure storage [Info](#)

Advanced

1x GiB ▼ Root volume, Not encrypted

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

Add new volume

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

File systems

▼ Summary

Number of instances [Info](#)

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 30 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't

- Suppose if you are going to buy a laptop, they will ask you right that what is the storage Requirement like One TB or 512 GB etc.
- Same way here.
- It will provide you 30 GB volume.
- Now this volume is actually root volume.
- It means it will install operating system inside this volume.
- We can add new volume from here also by clicking ADD new volume but we don't do this now.
- Now simply scroll down.

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 30 GiB

Cancel

Launch instance

Preview code

- And click Launch instance.

Instances (1) [Info](#)

Last updated [Refresh](#) less than a minute ago

Connect

Instance state ▼

Actions ▼

Launch instances ▼

Saved filter sets

Choose filter set ▼

Find Instance by attribute or tag (case-sensitive)

< 1 >

Settings

☐	Name ↗ ▼	Instance ID	Instance state ▼	Instance type ▼	Status check	Availabili
☐	WindowsInsta...	i-0790663b2d2e53b9b	Running ↗ ↻	t3.micro	Initializing	ap-south-

- Now you see our instance is created but current status of our instance is initializing so wait for status check become 3/3 checks passed

Instances (1/1) [Info](#)

Last updated
less than a minute ago

[Connect](#)[Instance state ▼](#)[Actions ▼](#)[Launch instances ▼](#)

Saved filter sets

[Choose filter set ▼](#)

Find Instance by attribute or tag (case-sensitive)

< 1 >



☒	Name	Instance ID	Instance state ▼	Instance type ▼	Status check	Availability
☒	WindowsInsta...	i-0790663b2d2e53b9b	Running	t3.micro	3/3 checks passed	ap-south-

i-0790663b2d2e53b9b (WindowsInstance)

[Details](#)[Status and alarms](#)[Monitoring](#)[Security](#)[Networking](#)[Storage](#)[Tags](#)

▼ Instance summary [Info](#)

Instance ID

i-0790663b2d2e53b9b

Public IPv4 address

13.232.250.79 | [open address](#)

Private IPv4 addresses

172.31.45.220

- Now see our status check becomes 3/3 checks passed.
- Now connect with our Machine.
- Select the created instance click on connect.

Connect [Info](#)

Connect to an instance using the browser-based client.

Instance ID

i-0790663b2d2e53b9b
(WindowsInstance)

VPC ID

vpc-072c4acd761c3b942

Security groups

sg-06110383e378abc30 (launch-wizard-277)

IAM role

—

[SSM Session Manager](#)[RDP client](#)[EC2 serial console](#)

- Now click RDP client.
- And scroll down.

Instance ID

i-0790663b2d2e53b9b (WindowsInstance)

Connection Type

☒ Connect using RDP client

Download a file to use with your RDP client and retrieve your password.

☐ Connect using Fleet Manager

To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and running on the instance. For more information, see [Working with SSM Agent](#)

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

[Download remote desktop file](#)

When prompted, connect to your instance using the following username and password:

Public DNS

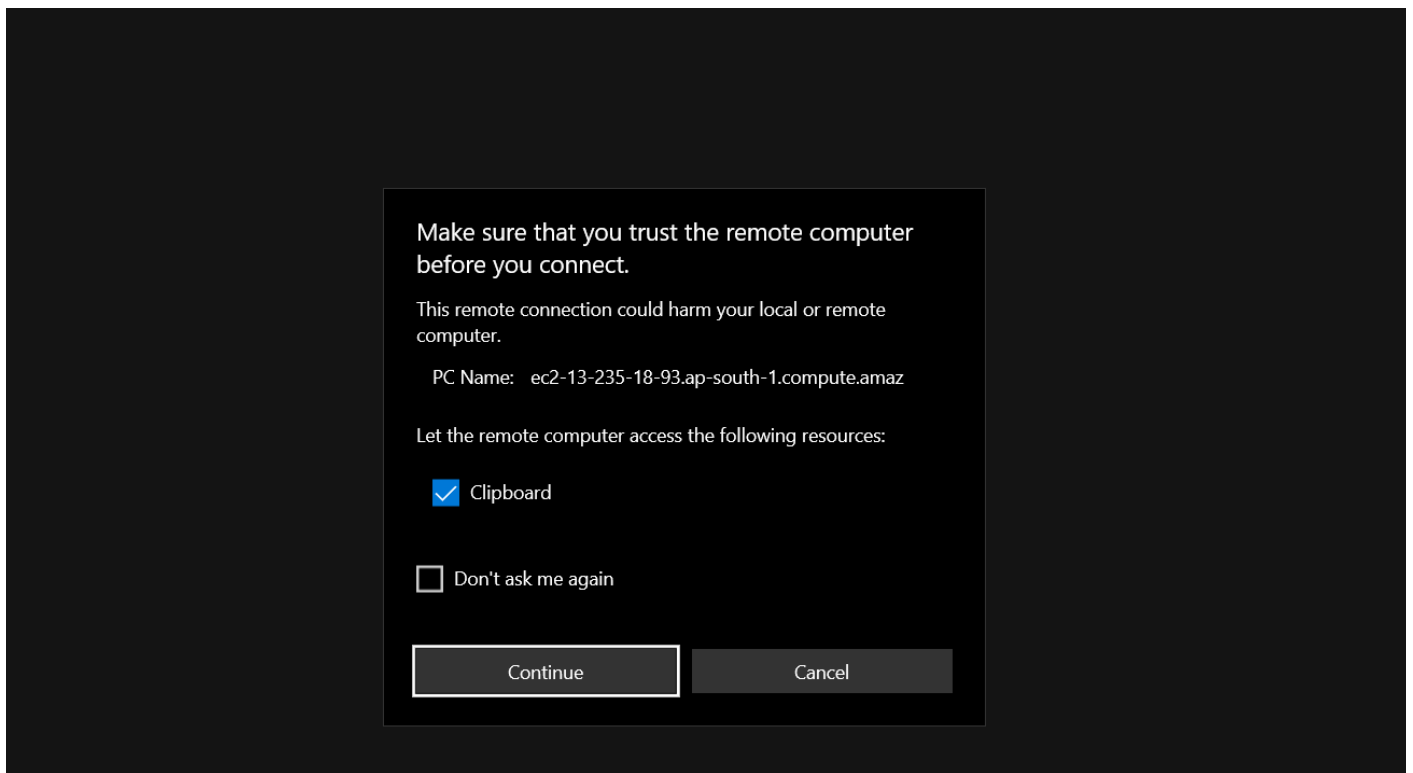
ec2-13-232-250-79.ap-south-1.compute.amazonaws.com

Username [Info](#)

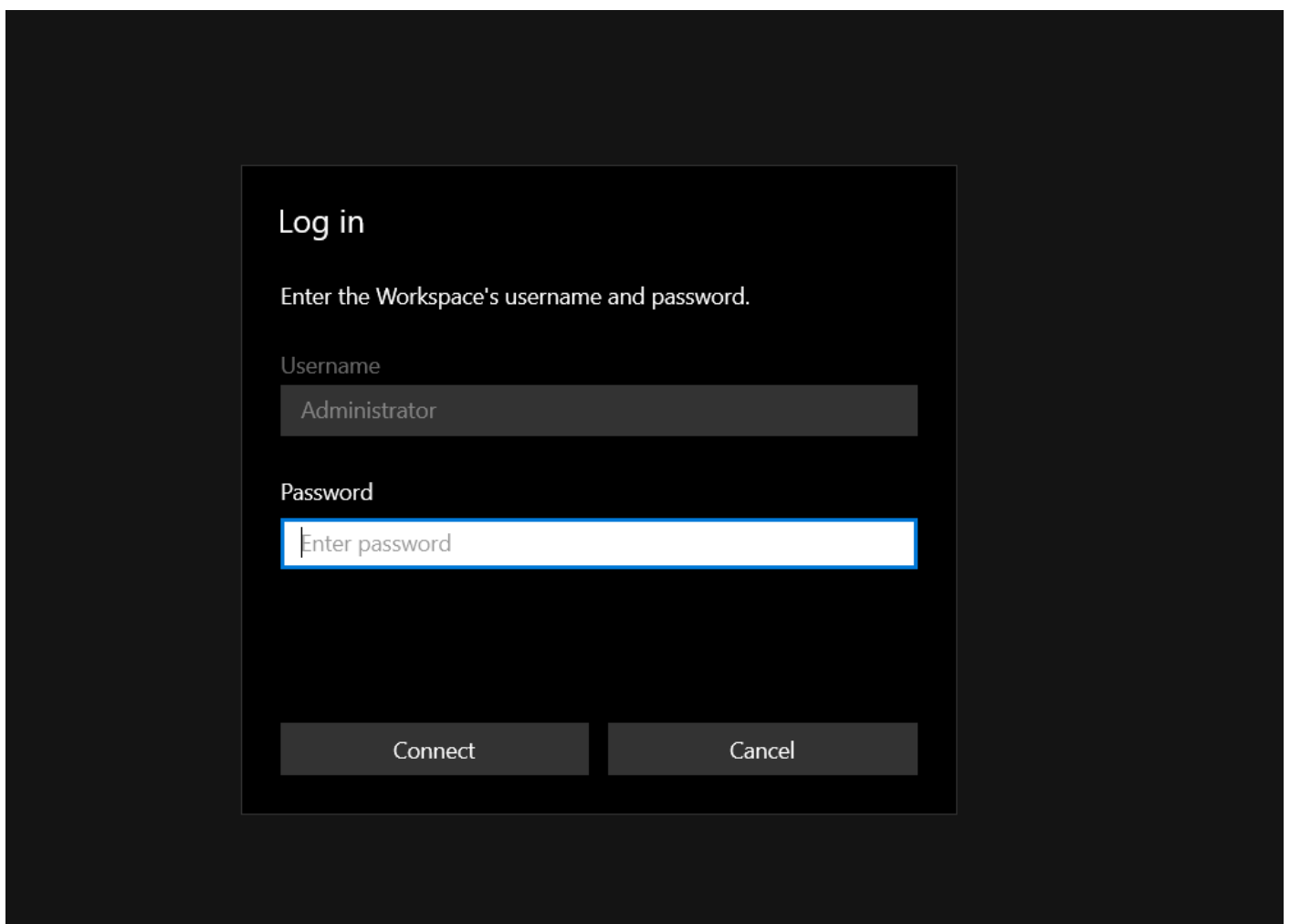
Administrator ▼

Password [Get password](#)

- Now click on download remote desktop file.
- When your file is downloaded go to download folder and open it.



- Now click continue.



- Now we need a password.
- Remember you created a demo.pem key in starting now we use it.
- Go to your browser instance tab.

Instance ID

 i-0790663b2d2e53b9b (WindowsInstance)

Connection Type

- ☒ Connect using RDP client
Download a file to use with your RDP client and retrieve your password.

☐ Connect using Fleet Manager
To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and running on the instance. For more information, see [Working with SSM Agent](#) 

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

 [Download remote desktop file](#)

When prompted, connect to your instance using the following username and password:

Public DNS

 ec2-13-232-250-79.ap-south-1.compute.amazonaws.com

Username [Info](#)

 Administrator ▼

Password [Get password](#)

- Now click on get password.

Get Windows password [Info](#)

Use your private key to retrieve and decrypt the initial Windows administrator password for this instance.

Instance ID

 i-0790663b2d2e53b9b (WindowsInstance)

Key pair associated with this instance

 demo

Private key

Either upload your private key file or copy and paste its contents into the field below.

 [Upload private key file](#)

Private key contents

- Now click on upload private key file.
- So you upload your demo.pem file.

 [Upload private key file](#)

demo.pem

1.67 KB

Private key contents

```
-----BEGIN RSA PRIVATE KEY-----
MIIEowIBAAKCAQEAvv236ltS1zMgilPBjRBrK5qInIJSLXDmX9bdpc6O+jrL5+t2
HcGPhXN/Doy3yaFD2Tm5N8OUpcNv6vh4LWBP7RBXe59bQgLeplIZ+Bsb+fQ02oAw
kfelEBxHC+5Rpz8Dm92rsOJxhfw/8+75K1wD4ywRhrtSRKcqZu4LxsWmE2HCoEhZ
eWn8yE/mmgCUIv1XjsfthocEBvcy1RK63RB+F58SyyXa7XudRzqZM46k9xncurFv
MxKfZHT+e/diwVDrAH/Ggt2+yep4U5LxqUK2WrusG0PnkOFFxAn8P6igS+Jr5ru
KS750veZ/pROghlmhLO2qUL46VHXzMnGjFlgQwIDAQABAolBAGFvWpZKxLZYtO8c
z49B0wr5qVv/DAbso/qcyTK/cB2kRUypz3Zauvdf9p/blh9dEBoP4rfaJULzrPL4
```

[Cancel](#)

[Decrypt password](#)

- Now simply scroll and click Decrypt password.

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

 [Download remote desktop file](#)

When prompted, connect to your instance using the following username and password:

Public DNS

 ec2-13-235-18-93.ap-south-1.compute.amazonaws.com

Username [Info](#)

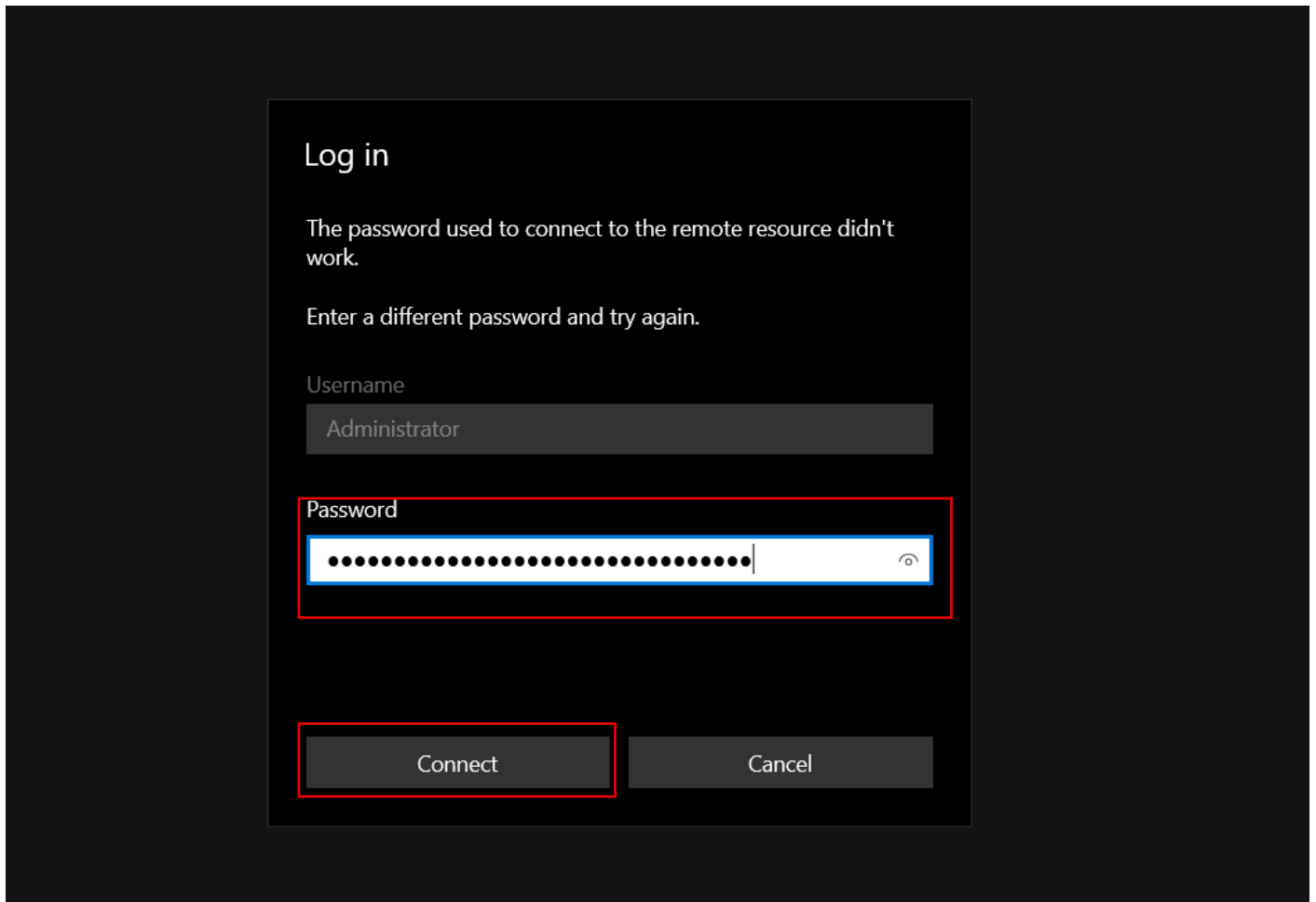
 Administrator ▼

Password

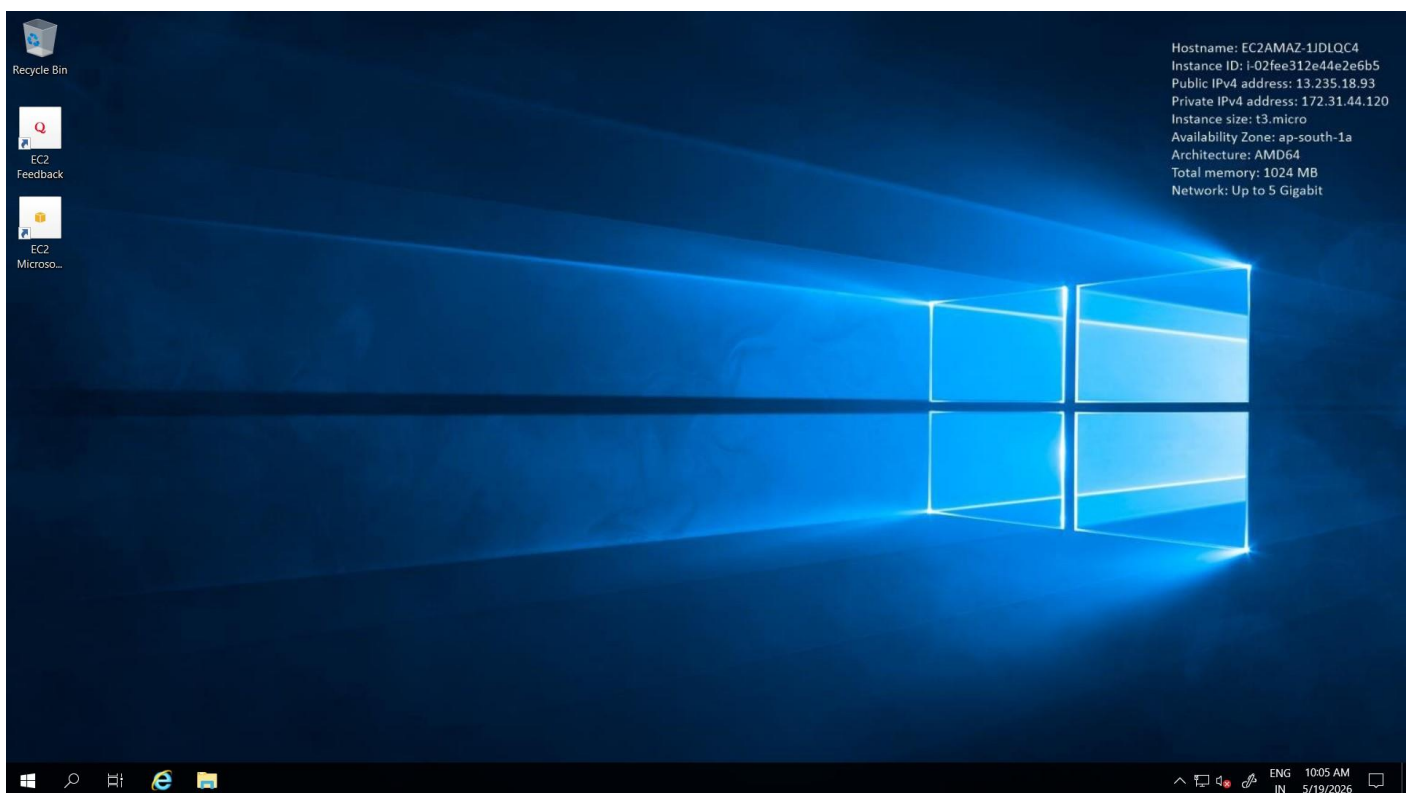
 @JNfy?%;z=wJhuaaavg;!%ukFy;gi8Dv

 If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

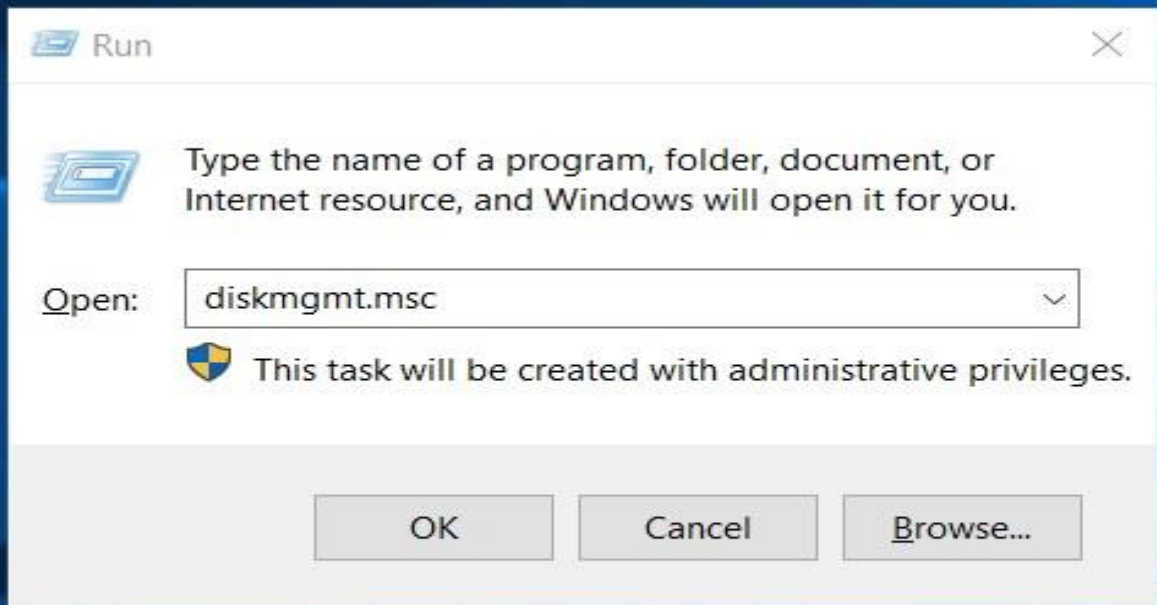
- Now simply copy the Password.



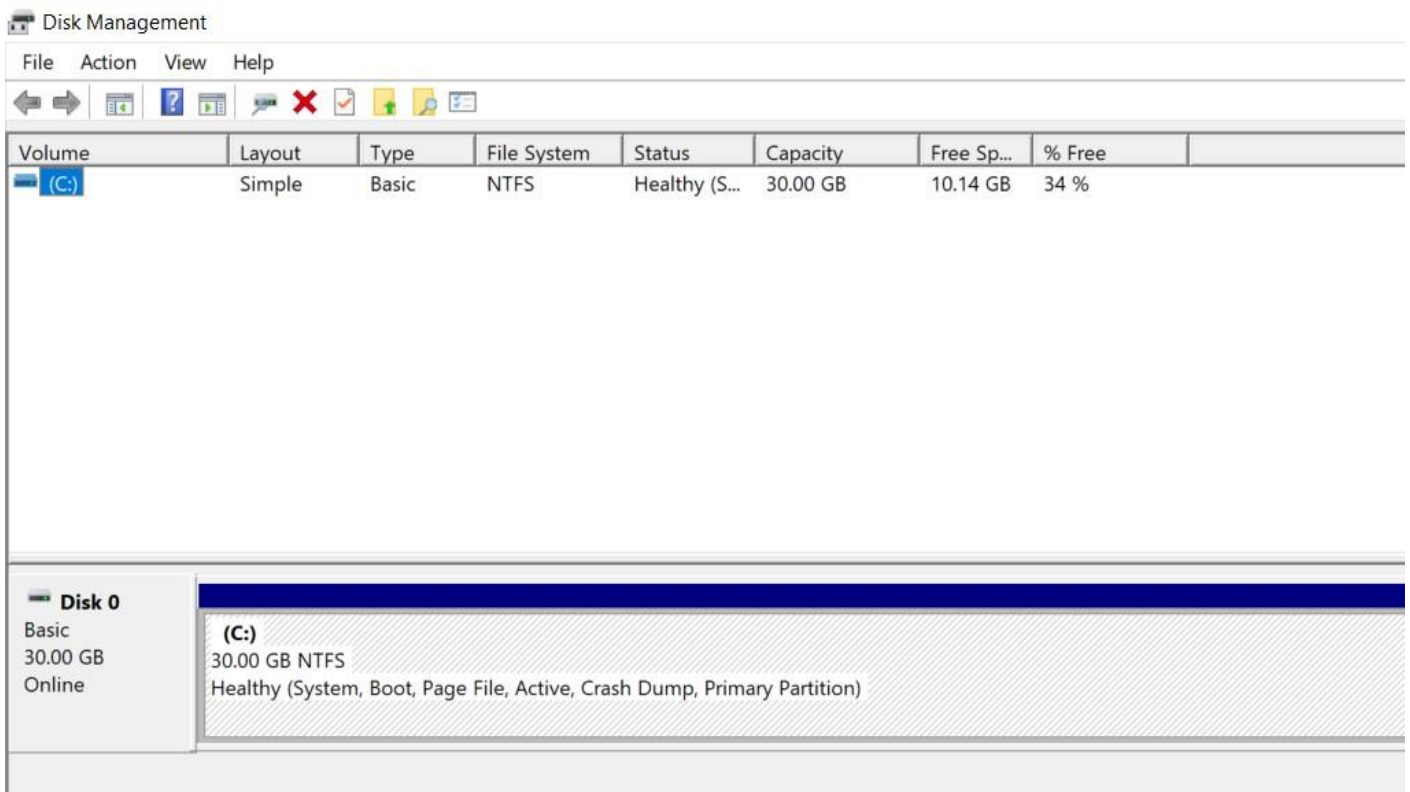
- Now paste the password in remote Desktop and click connect.



- Now you see this is the window machine currently we created.
- Now just press windows key and r



- Then type diskmgmt.msc in run terminal.



- Now here I have 30 GB.
- Now I know that this 30 GB volume is my root drive and in normal circumstances we are not storing data over here.
- So, I want to add additional volume.
- Go back to your EC2 tab in your browser.

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

▼ Network & Security

Security Groups

Elastic IPs

Placement Groups

Instances (1/1) Info

Last updated 24 minutes ago

Connect

Instance state ▼

Actions ▼

Launch instances ▼

Saved filter sets

Choose filter set ▼

Find Instance by attribute or tag (case-sensitive)

☒	Name ↗	Instance ID	Instance state	Instance type	Status check	Availability
☒	WindowsInsta...	i-0790663b2d2e53b9b	Running ↗ ↗	t3.micro	3/3 checks passed	ap-south-

- Now in nav bar inside Elastic Block Store click Volumes.

Volumes (1) Info

Last updated less than a minute ago

Recycle Bin

Actions ▼

Create volume

Saved filter sets

Choose filter set ▼

Search

☐	Name ↗	Volume ID	Type	Size	IOPS	Throughput	Snapsho
☐		vol-09dfa606973676fa9	gp2	30 GiB	100	-	snap-0d9

- Now simply click Create volume.

Volume settings

Volume type [Info](#)

General Purpose SSD (gp3) ▼

Size (GiB) [Info](#)

15

Min: 1 GiB, Max: 65536 GiB.

IOPS [Info](#)

3000

Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s) [Info](#)

125

Min: 125 MiB, Max: 2000 MiB. Baseline: 125 MiB/s.

- Volume type General Purpose SSD (gp3)
- Size 15gb

Availability Zone [Info](#)

aps1-az1 (ap-south-1a) ▼

Snapshot ID - optional [Info](#)

Don't create volume from a snapshot ▼



Encryption [Info](#)

Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.

☐ Encrypt this volume

- Availability Zone ap-south-1a .
- It's most important our storage and Ec2 instance have same Availability Zone.

Snapshot summary [Info](#)

Click refresh to view backup information

The volume type that you select and the tags that you assign determine whether the volume will be backed up by any Data Lifecycle Manager policies.

Cancel [Create volume](#)

- Click Create volume.

Volumes (2) [Info](#)

Last updated less than a minute ago

[Recycle Bin](#) [Actions](#) [Create volume](#)

Saved filter sets [Choose filter set](#)

☐	Name ✎	Volume ID	Type	Size	IOPS	Throughput	Snapshot
☐		vol-09dfa606973676fa9	gp2	30 GiB	100	-	snap-0d5
☐		vol-056c64b7960a2ef74	gp3	15 GiB	3000	125	-

- Now we have two volume.

Volumes (1/2) [Info](#)

Last updated less than a minute ago

[Recycle Bin](#) [Actions](#) [Create volume](#)

Saved filter sets [Choose filter set](#)

☐	Name ✎	Volume ID	Type	Size
☐		vol-09dfa606973676fa9	gp2	30 GiB
☒		vol-056c64b7960a2ef74	gp3	15 GiB

Volume ID: vol-056c64b7960a2ef74

[Details](#) [Status checks](#) [Monitoring](#) [Tags](#)

Volume ID	Size	Type
vol-056c64b7960a2ef74	15 GiB	gp3

- Modify volume
- Create snapshot
- Create snapshot lifecycle policy
- Delete volume
- Attach volume**
- Detach volume
- Force detach volume
- Manage auto-enabled I/O
- Copy volume - new
- Manage tags
- Resilience testing

- Now select created volume and click Action and then select Attach volume.

Basic details

Volume ID

[vol-056c64b7960a2ef74](#)

Availability Zone

aps1-az1 (ap-south-1a)

Instance [Info](#)

i-0790663b2d2e53b9b (WindowsInstance) (running)

Only instances in the same Availability Zone as the selected volume are displayed.

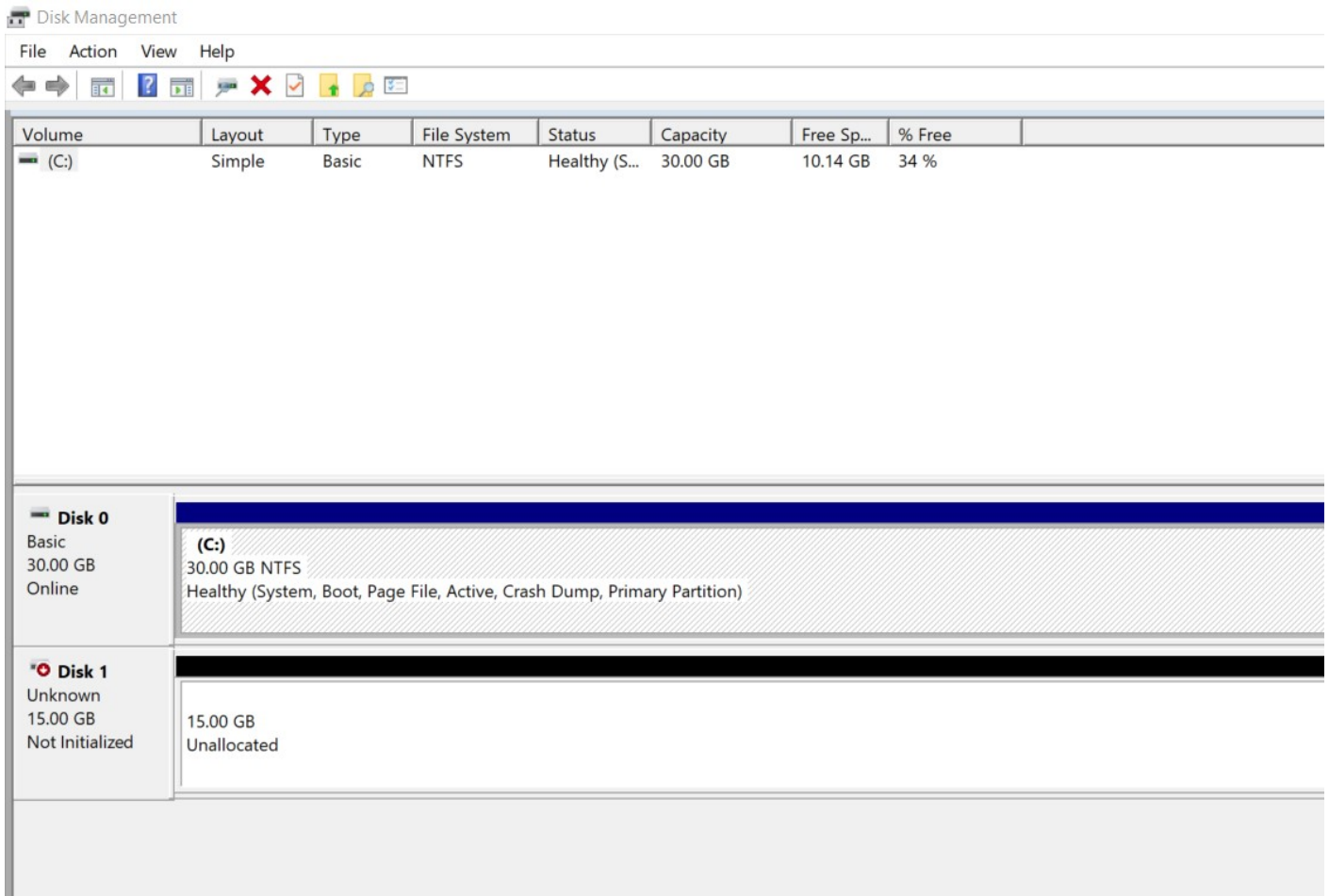
Device name [Info](#)

xvdf

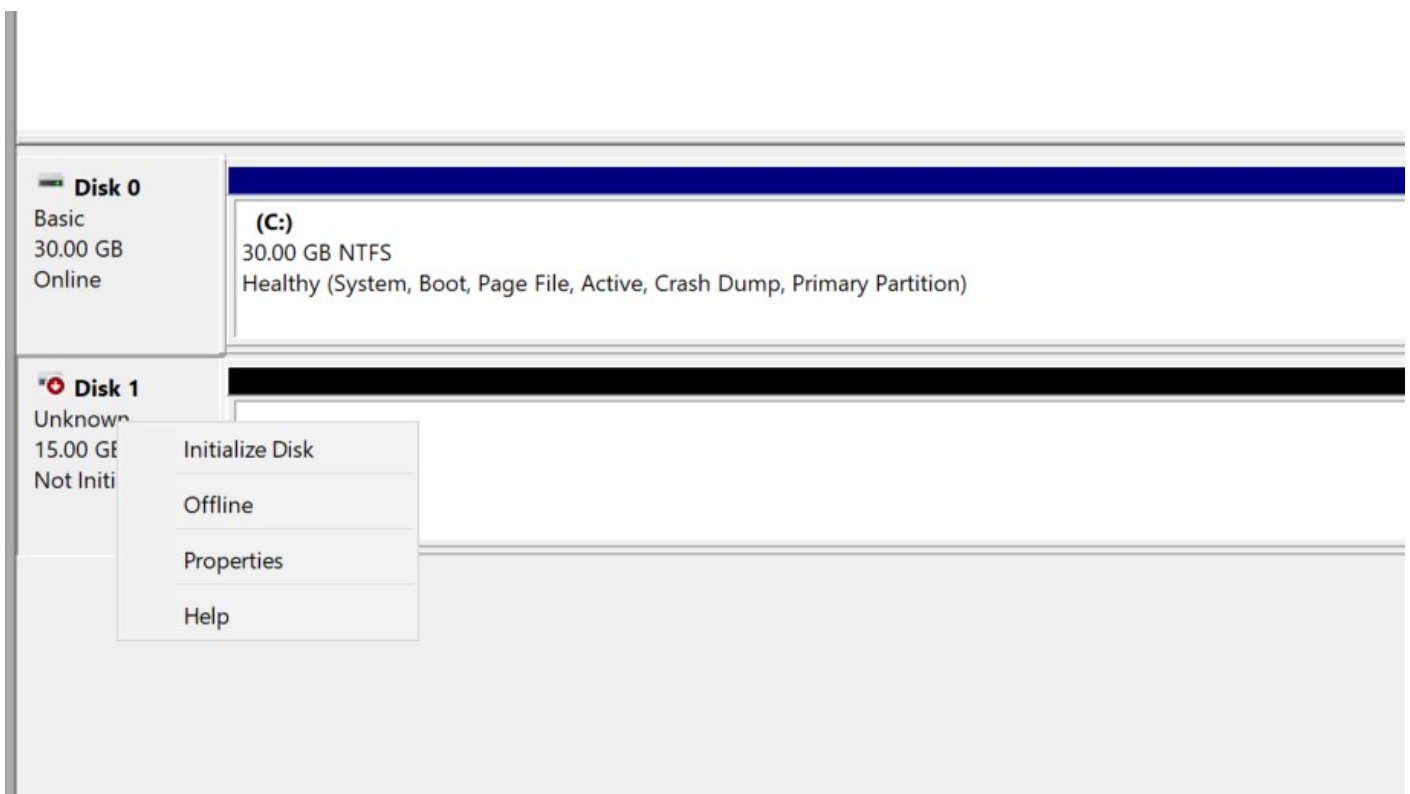
Recommended device names for Windows: /dev/sda1 for root volume. xvd[f-p] for data volumes.

Cancel [Attach volume](#)

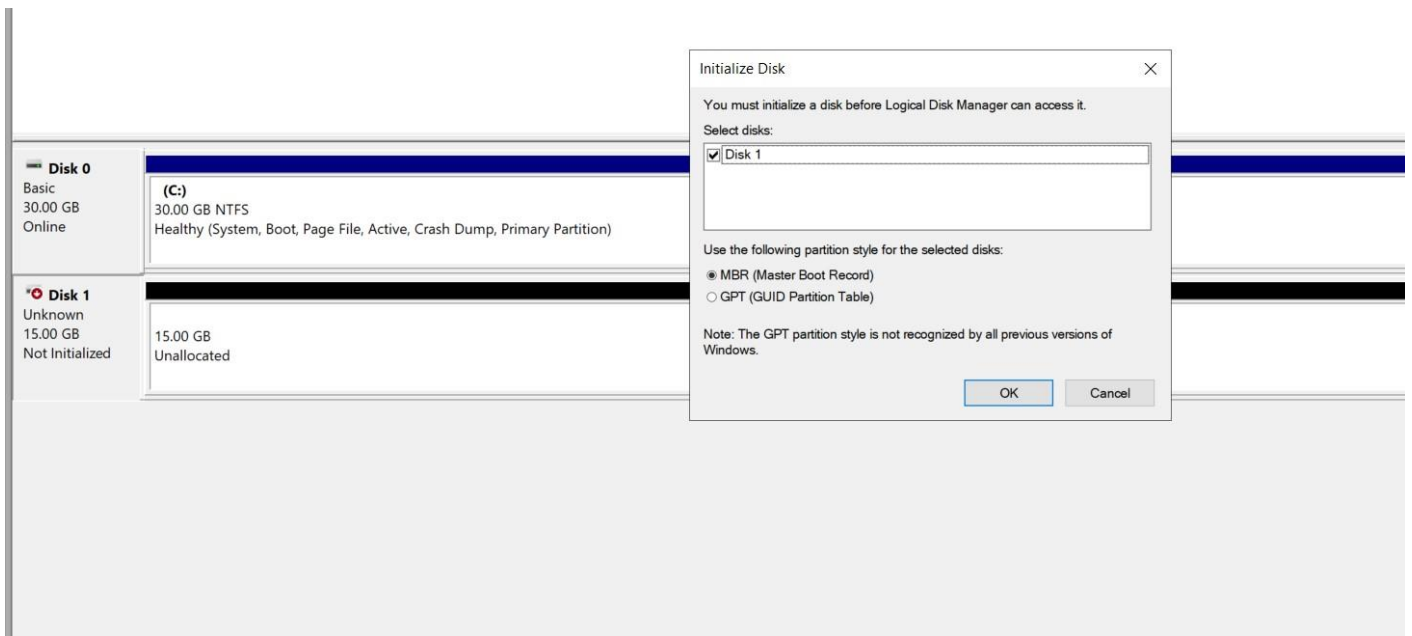
- Now select your Instance and device xvdf.
- And then click Attach volume.



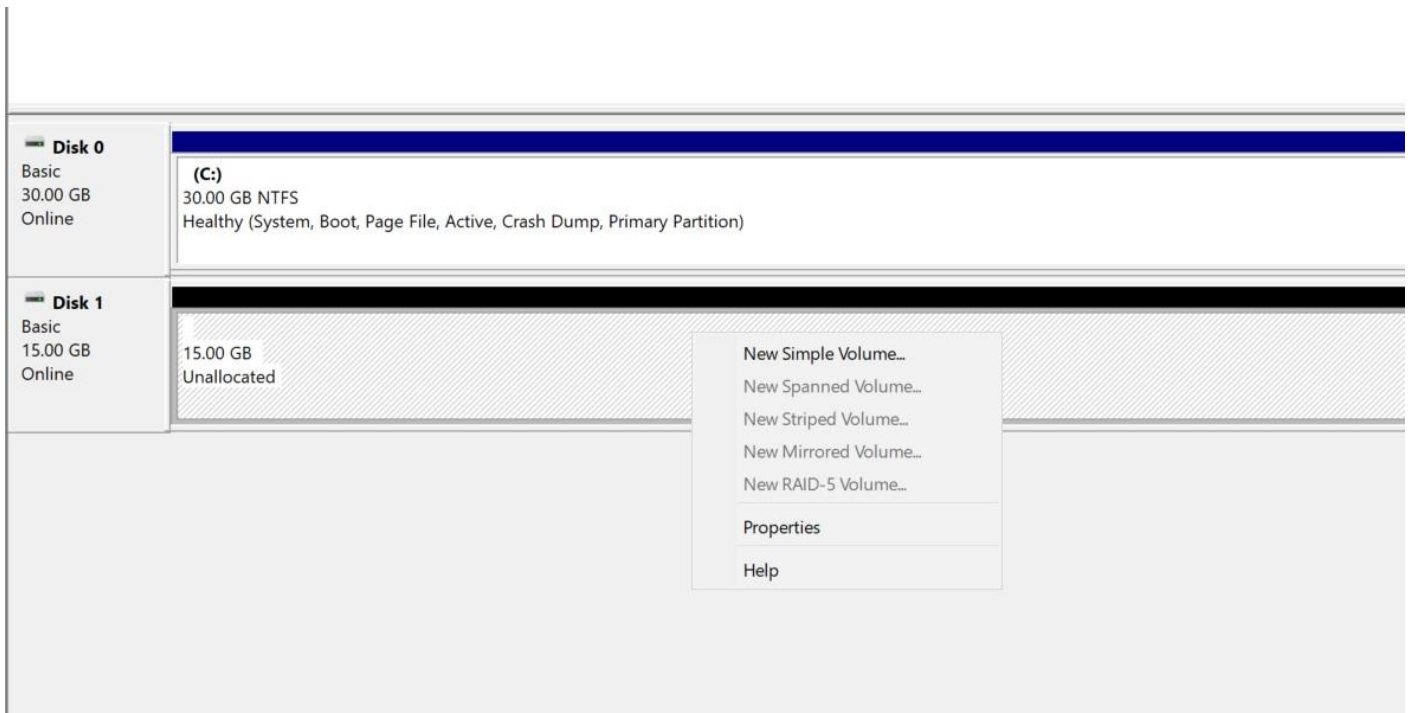
- Now this is added but you go to my pc its's not visible.



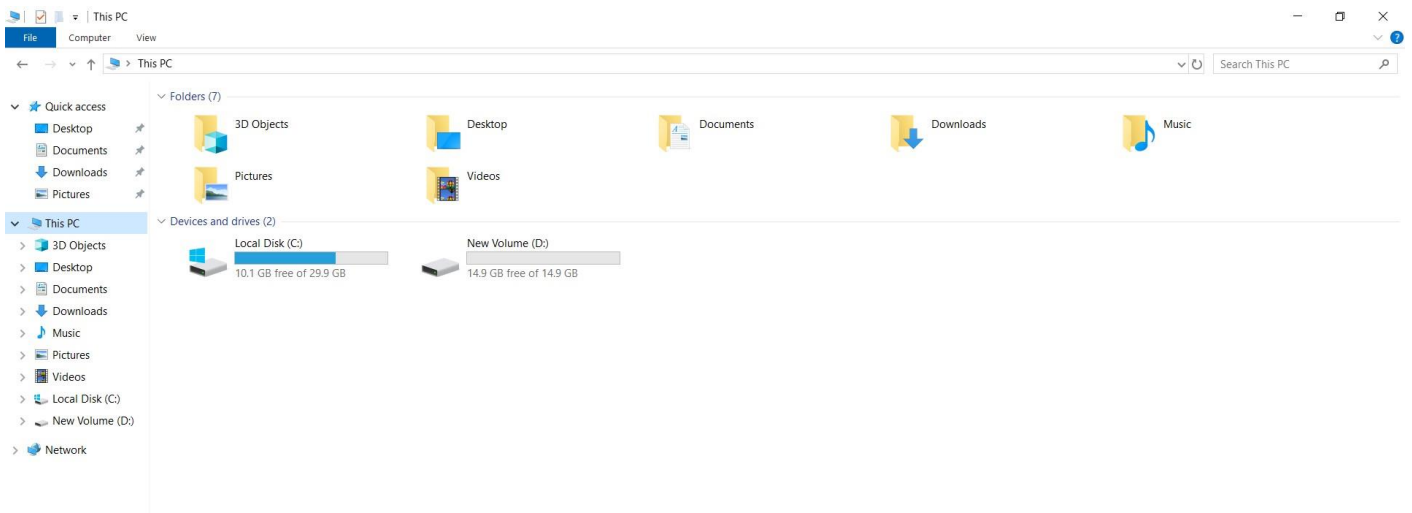
- For that this driver shows in my pc right click on it.
- Then select initialize Disk.



- Now simply click ok.



- Now right click on unallocated side and select New Simple Volume.
- Then click next. Next and last finish.
- All process is done now our new drive shows in this pc.
- So go inside This Pc.



- Now you see your new drive is visible in this pc.
- Now a scenario you need more storage.
- Then again go to volume tab in your browser in your real window.

Volumes (1/2) Info Last updated 14 minutes ago Recycle Bin Actions Create volume

Saved filter sets Choose filter set Search

	Name	Volume ID	Type	Size
☐		vol-09dfa606973676fa9	gp2	30 GiB
☒		vol-056c64b7960a2ef74	gp3	15 GiB

Volume ID: vol-056c64b7960a2ef74

Details	Status checks	Monitoring	Tags
Volume ID vol-056c64b7960a2ef74	Size 15 GiB	Type gp3	
AWS Compute Optimizer finding	Volume state In-use	IOPS 3000	Throughput 125

Modify volume

- Create snapshot
- Create snapshot lifecycle policy
- Delete volume
- Attach volume
- Detach volume
- Force detach volume
- Manage auto-enabled I/O
- Copy volume - new
- Manage tags
- Resilience testing

- Select your created volume and click Action.
- And select Modify volume.

Volume details

Volume ID
 vol-056c64b7960a2ef74

Volume type | Info
 General Purpose SSD (gp3)

Size (GiB) | Info

 Min: 1 GiB, Max: 65536 GiB.

IOPS | Info

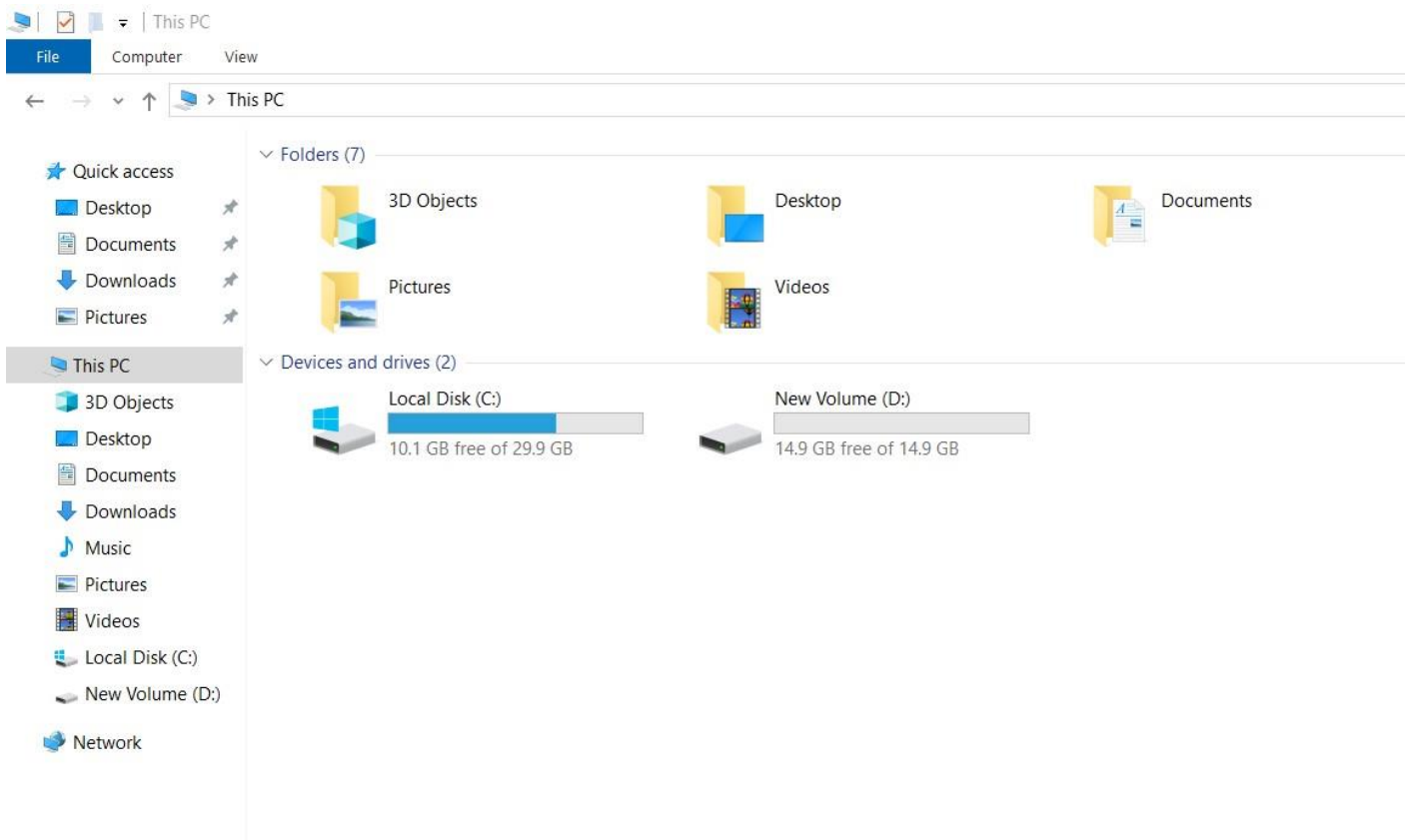
 Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s) | Info

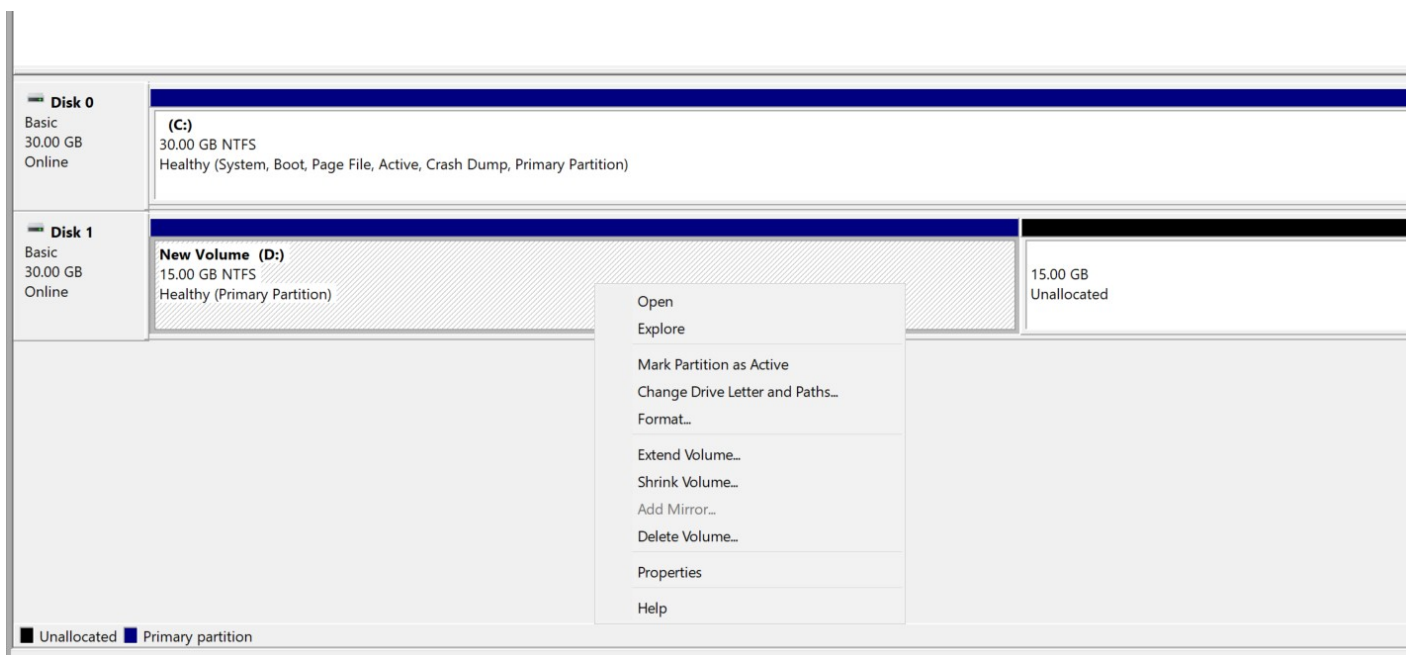
 Min: 125 MiB, Max: 2000 MiB, Baseline: 125 MiB/s.

Cancel Modify

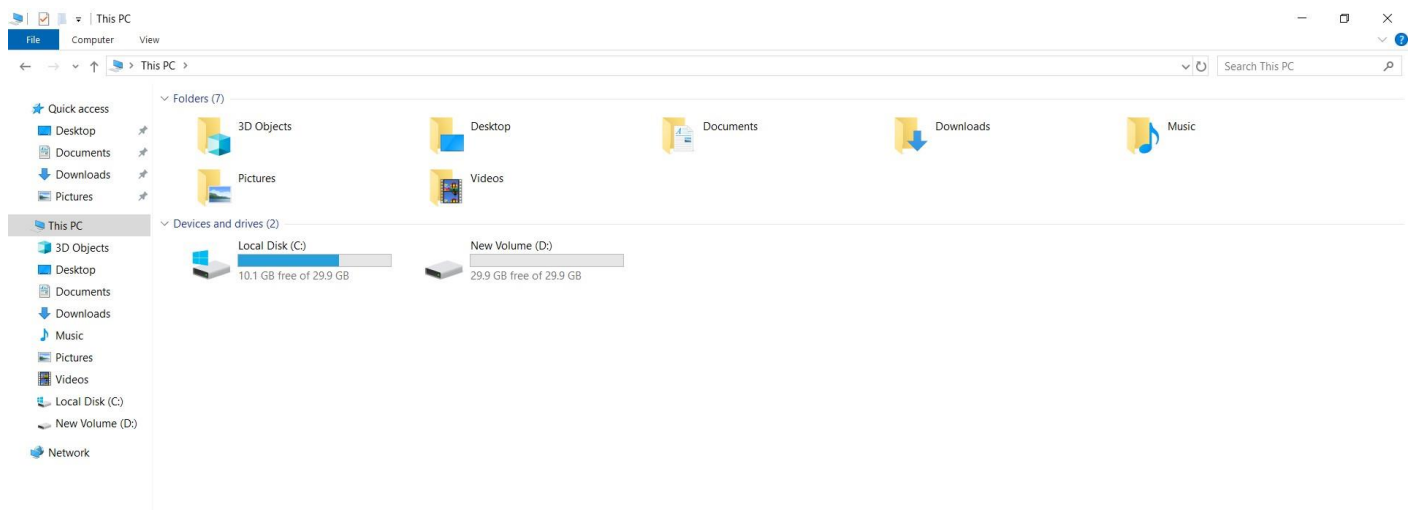
- Now increase it to 30gb and click Modify.



- Now see our new drive storage is 14.9gb not 30.
- So again press windows key and r then type diskmgmt.msc



- Our increased volume is present but not added.
- So right click on healthy new volume D and select Extend Volume.
- Then simply click next. Next and finish.
- And close this window and open this pc.



- Now see our disk volume is increased to 29.9 gb.